Manual for the FTSE Stock Clustering Notebooks

Introduction

This project groups FTSE listed companies whose share price behaviour moves together. Three Jupyter notebooks apply different clustering methods (K-Means, Gaussian Mixture Models and K-Medoids with Dynamic Time Warping) to three standardised datasets of daily stock values. Each notebook cleans the data, compares several numbers of clusters k using validity scores and a consensus stability check, and then reports which stocks belong to which cluster. This page explains what each file holds and how to run the code.

What each file contains

File	What it contains
FTSE_100_2015_2025_Dataset.csv	103 FTSE 100 stocks by 2,779 trading days, from 2 Jan 2015 to 31 Dec 2025. It has 17 empty cells.
FTSE_250_2021_2026_Dataset.csv	241 FTSE 250 stocks by 1,153 trading days, from 30 Nov 2021 to 29 Jun 2026. It has 4 empty cells.
FTSE_250_2025_2026_Dataset.csv	250 FTSE 250 stocks by 273 trading days, from 2 Jun 2025 to 29 Jun 2026. It has 216 empty cells, so the cleaning step will drop some stocks.
kmeans_clustering.ipynb	15 cells. Reduces the data with PCA (90 percent of variance), then runs K-Means for k from 2 to 10. It scores each k with silhouette, Calinski-Harabasz, Davies-Bouldin and Dunn, checks stability with consensus matrices from 100 random subsamples (80 percent each), and lists the stocks in each cluster.
gmm_em_clustering.ipynb	18 cells. Applies PCA, then fits a Gaussian Mixture Model by the EM algorithm for k from 2 to 8 and four covariance types (spherical, diag, tied, full). It chooses the model with the lowest BIC, uses log-likelihood as a convergence check, adds consensus plots, and flags stocks with a membership probability below 0.70 as uncertain.
kmedoids_dtw_pam_clustering.ipynb	16 cells. Builds a Dynamic Time Warping distance matrix (window of 5 percent of the series length) with dtaidistance, then runs PAM K-Medoids for k from 2 to 10. It adds an MDS map and names a medoid ticker, the most representative stock, for each cluster. It is the heaviest to compute.

All three CSV files share one layout. The first column is Ticker (for example RSW.L), every other column header is a trading date, and each row holds the standardised values of one stock, with a mean near 0 and a standard deviation near 1.

How to work on the code

1. **Set up.** Use Google Colab. Open each notebook there and nothing needs to be installed.

2. **Choose a pair.** Pick one notebook and one CSV file, and keep them in the same folder. Use the same CSV in all three notebooks if you want to compare methods.

3. **Load the data.** In Colab, run the second cell and upload the CSV. On your own computer that cell only prints the file name, so first add a line such as FILENAME = "FTSE_100_2015_2025_Dataset.csv" to it (the GMM notebook uses DATA_FILE instead).

4. **Run every cell in order.** Use Run All, or run from the top downwards. Stocks with more than 10 percent missing values are dropped, smaller gaps are filled forward and then backward, and the printout lists what was removed.

5. **Read the results.** The metrics table and plots show how each k performs, the summary box gives the final scores and cluster sizes, and the last cell lists the tickers in every cluster.

6. **Tune if needed.** Near the top you can change K_RANGE, VARIANCE_TARGET, MAX_MISSING_FRAC and N_SUBSAMPLES. In the K-Means and K-Medoids notebooks you can also set FINAL_K by hand to override the suggested k. Keep RANDOM_STATE at 42 for repeatable results and rerun from the first cell after any change.

7. **Save your output.** The notebooks do not write results to disk, so export the cluster assignments yourself, for example with the pandas to_csv method, before closing the session.